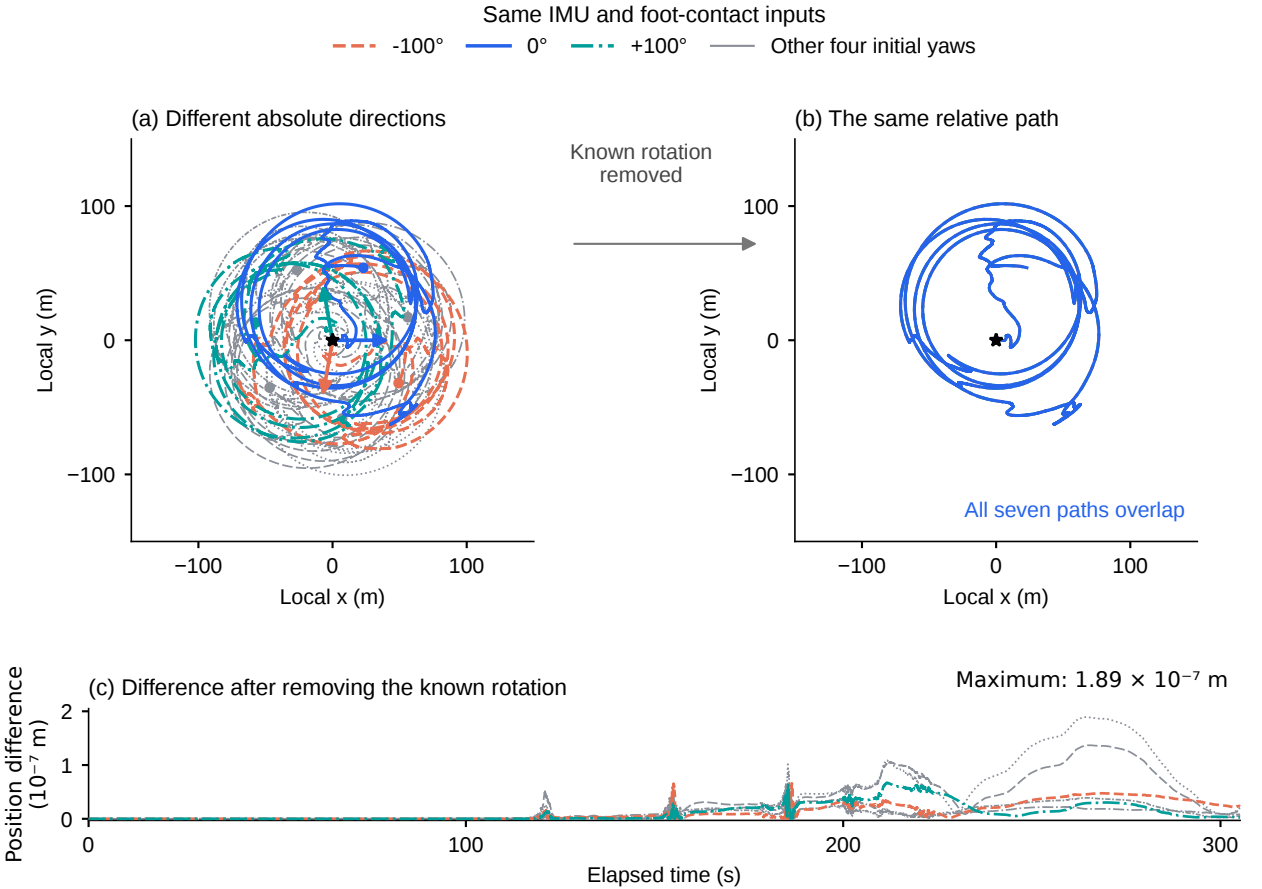

Same IMU and foot-contact inputs
-100°
0°
+100°
Other four initial yaws
(a) Different absolute directions
Known rotation
removed
(b) The same relative path
Local y (m)
Local x (m)
100
0
−100
All seven paths overlap
(c) Difference after removing the known rotation
Maximum: 1.89 × 10⁻⁷ m
Position difference
(10⁻⁷ m)
Elapsed time (s)
0
1
2
200
300